

Passive Autonomy

FSAE Scaled Dynamics Validation Platform, Electrical Systems Documentation

Subsystem: Power Architecture (ESPCA)

Document Scope: Step 2 of the Electronics Development Pipeline

Document Information

Field	Detail
Project	Passive Autonomy Unified Control Framework for FSAE
Validation Platform	1/5–1/10 Scale RC Vehicle (Scaled Dynamics Validation Platform)
Core Electronics	Raspberry Pi Pico (MCU), Sipeed Tang Nano 20K (FPGA), ESP32 (Telemetry)
Battery	2S LiPo, 7.4V Nominal (6.0V–8.4V Operating Range)
Subsystem Covered	Power Architecture ESPCA (Electrical Segmentation and Power Control Architecture)
Status	Simulation Complete Validated in LTspice

Introduction

This document is part of the ongoing electronics development effort for the Passive Autonomy project, a control framework designed to close the bandwidth gap between human reaction time and the physical dynamics of an FSAE race vehicle. Before implementation on the full-scale competition car, every subsystem is first validated on a scaled RC platform, referred to as the Scaled Dynamics Validation Platform.

The electrical development of the validation platform is organized into eleven sequential steps, each building on the outputs of the last. The table below summarizes the full roadmap and indicates which step this document covers.

Step	Title	Status
1	System Architecture & Requirements	Completed
2	Power Architecture (ESPCA)	Completed This Document
3	Sensor Selection & Interfacing	Planned
4	Core MCU Firmware Skeleton	Planned
5	Control Algorithm Design (AIS v3 / Gyro Estimation)	Planned
6	FPGA Layer	Planned
7	Communication Architecture	Planned
8	PCB Design	Planned
9	Bring-Up & Hardware Validation	Planned
10	Firmware / FPGA Integration on RC Chassis	Planned
11	Data Logging & Validation	Planned

Power Architecture (ESPCA)

1.1 Objective

The goal of this step is to design a power system in which high-current motor switching noise cannot corrupt the sensor and logic rails that the AIS v3 control law depends on. This is the electrical implementation of the ESPCA (Electrical Segmentation and Power Control Architecture) principle defined in the Passive Autonomy framework: physically and electrically segmented power domains, with a single-point star ground and filtering at every domain boundary. Without a clean sensor supply, the high-pass vibration filter at the heart of AIS v3 cannot distinguish real mechanical vibration from electrical noise.

1.2 System Architecture Overview

The power chain steps down the main battery voltage in two stages: a switching buck converter produces an intermediate 5V logic rail for the three main compute boards, and a linear low-dropout regulator further steps this down to a clean, low-noise 3.3V rail dedicated to the IMU and other sensors. Each board's own onboard regulation is left untouched, the 5V and 3.3V rails supply board-level inputs, not internal chip rails.

1.3 Logic Rail Current Budget

Current draw for every board and sensor was estimated from manufacturer datasheets, then summed with a 50% design margin to size the buck converter and confirm headroom.

Device	Current Draw	Basis
Raspberry Pi Pico (MCU)	100 mA	Board-level estimate, incl. onboard LDO
Sipeed Tang Nano 20K (FPGA)	500 mA	Manufacturer rated demand (5V, USB-C)
ESP32 (Telemetry)	250 mA	Typical active draw with Wi-Fi enabled
ICM-20948 IMU	5 mA	Datasheet operating current (3.11 mA), rounded up
ACS758 Current Sensor	15 mA	Typical Hall-effect sensor supply current
A3144 Hall Sensors (×2)	20 mA	10 mA each, wheel speed sensing
Subtotal	890 mA	—
+ 50% Margin	445 mA	—
Design Target	≈ 1.5 A	Rounded for component selection

1.4 Buck Converter [LT8610]

The LT8610 was selected for its wide input range, synchronous topology, and low-EMI design, all directly relevant to keeping the downstream sensor rail clean.

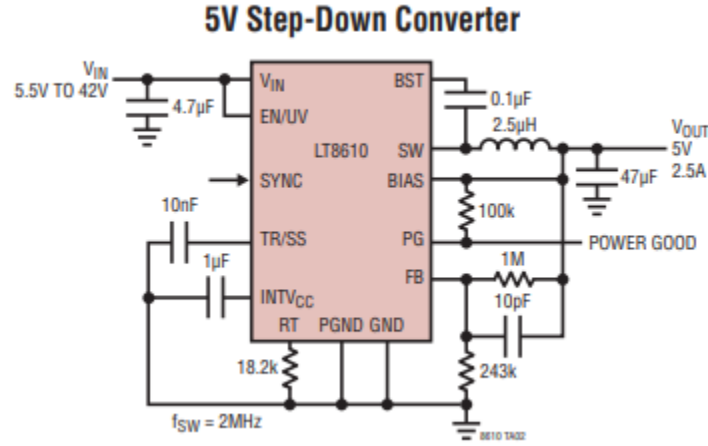


Figure 1: Datasheet Reference for the LT8610.

Pin / Component	Value	Function
V1 (Battery)	7.4 V nominal (6.0–8.4 V)	2S LiPo input
C1 (Cin)	4.7 μ F	Input decoupling
L1 (Inductor)	2.5 μ H	Switching energy storage
C5 (Cout)	47 μ F	Output filtering
R1 / R2 (FB Divider)	243 k Ω / 18.2 k Ω	Sets 5V output via 0.970V reference
C6 (Phase-Lead)	10 pF	FB loop stability
R4 (RT)	1 M Ω	Sets switching frequency
C2 (TR/SS)	10 nF	Soft-start timing
C3 (INTVcc)	1 μ F	Internal rail decoupling
C4 (BST)	0.1 μ F	High-side gate drive bootstrap
R3 (BIAS)	100 k Ω to GND	Internal LDO bias reference
EN/UV	Tied to VIN	Always-enabled operation
SYNC	Tied to GND	Free-running (Burst Mode) switching

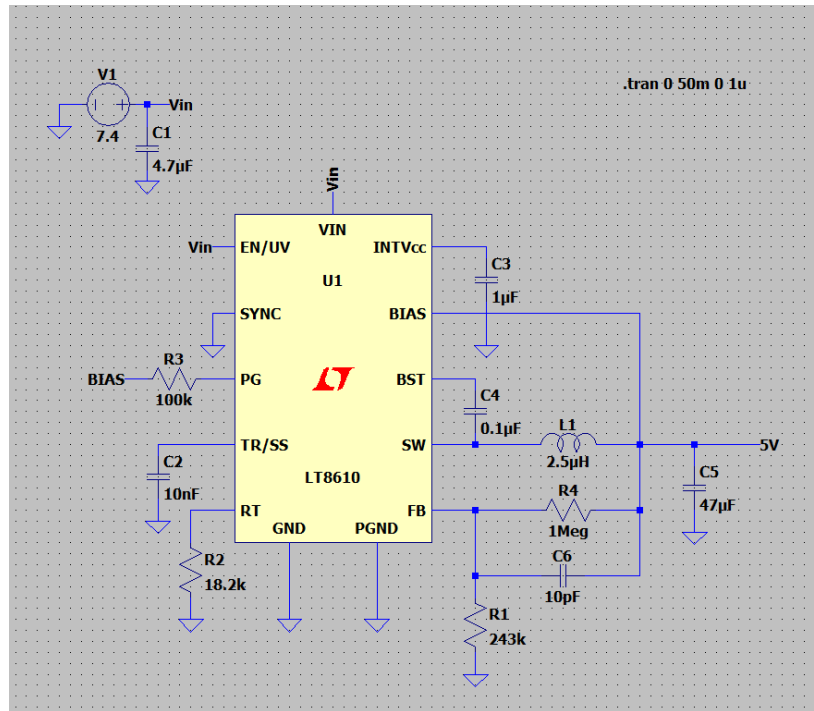


Figure 1.1: LTspice build of the LT8610 buck converter.

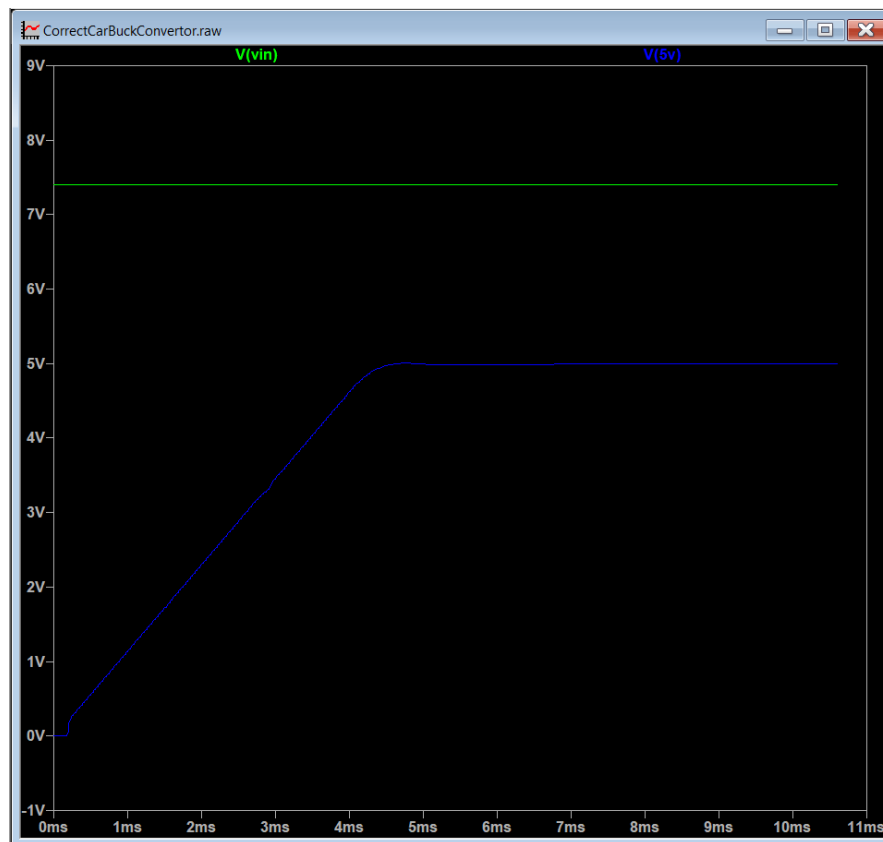


Figure 1.2: $V(vin)$ and $V(vout)$ transient showing a clear stable 5 volts output.

1.5 LDO LT1761 / LT1762 3.3V Regulator

The 3.3V sensor rail uses the LT1761/LT1762 series, a low-noise, micropower LDO from the same manufacturer as the LT8610. With the bypass capacitor populated, the datasheet specifies 20 μVRMS output noise (10Hz–100kHz), well suited to feeding the IMU.

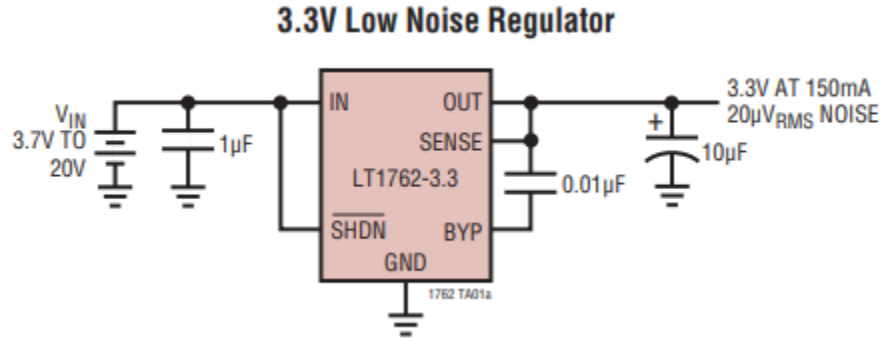


Figure 1.3: Datasheet Reference for the LT1762.

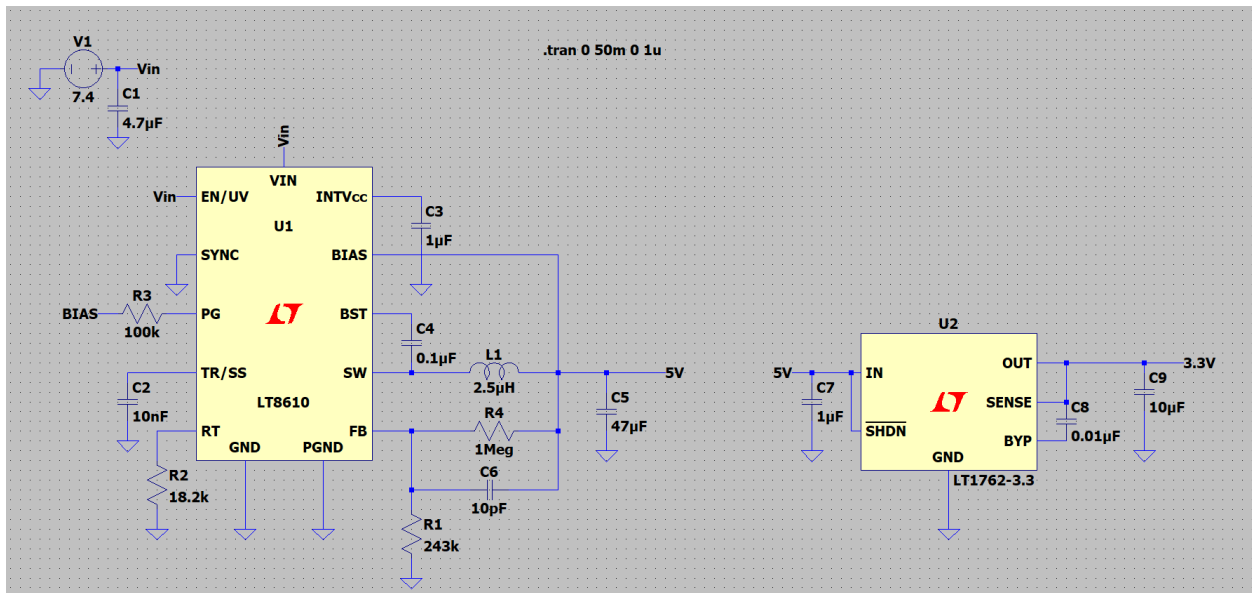


Figure 1.4: Combined buck + LDO schematic.

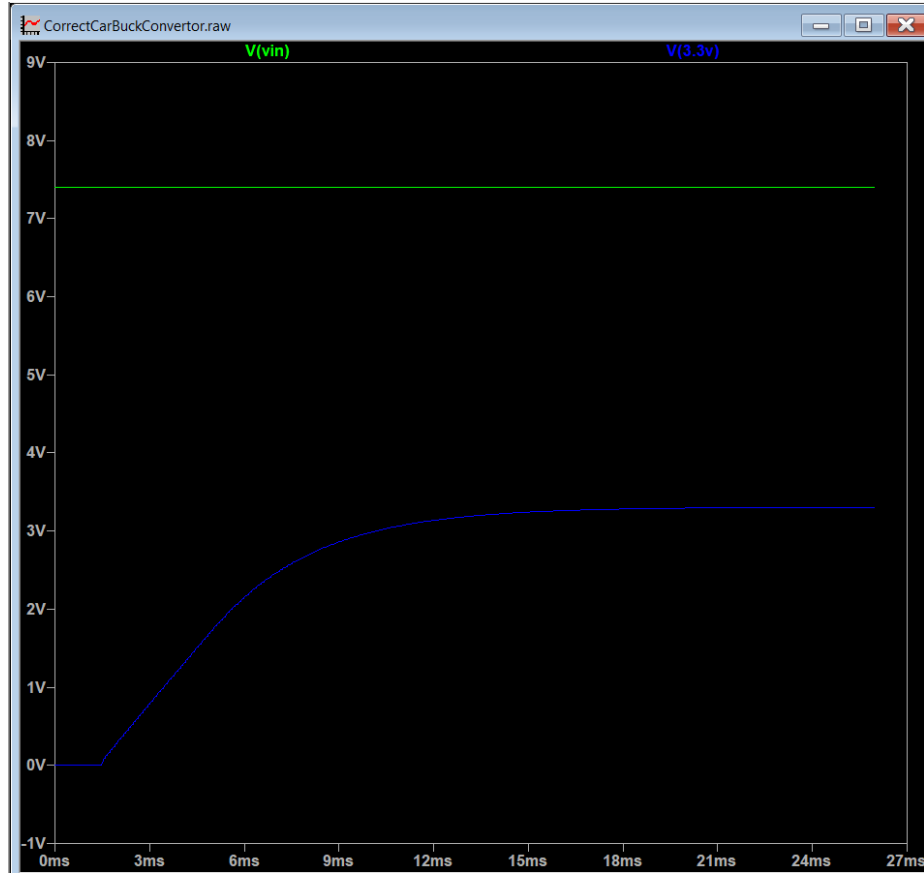


Figure 1.5: $V(vin)$ and $V(vout)$ transient showing a clear stable 3.3 volts output.

1.6 Final Validated Values

Stage	Parameter	Final Value
Buck (LT8610)	Output Voltage	$\approx 4.97\text{ V}$
Buck (LT8610)	Inductor (L1)	$2.5\text{ }\mu\text{H}$
Buck (LT8610)	Output Cap (C5)	$47\text{ }\mu\text{F}$
Buck (LT8610)	FB Divider (R1 / R2)	$243\text{ k}\Omega / 18.2\text{ k}\Omega$
Buck (LT8610)	RT (R4)	$1\text{ M}\Omega$
LDO (LT1761/1762)	Output Voltage	3.3 V
LDO (LT1761/1762)	Output Cap (C9)	$1\text{ }\mu\text{F}$
LDO (LT1761/1762)	BYP Cap	$0.01\text{ }\mu\text{F}$
System	Design Current Target	$\approx 1.5\text{ A}$ (5V rail)

1.7 KiCad Implementation

The schematic done in LTspice was then implemented in KiCad for footprint assignment and PCB Design.

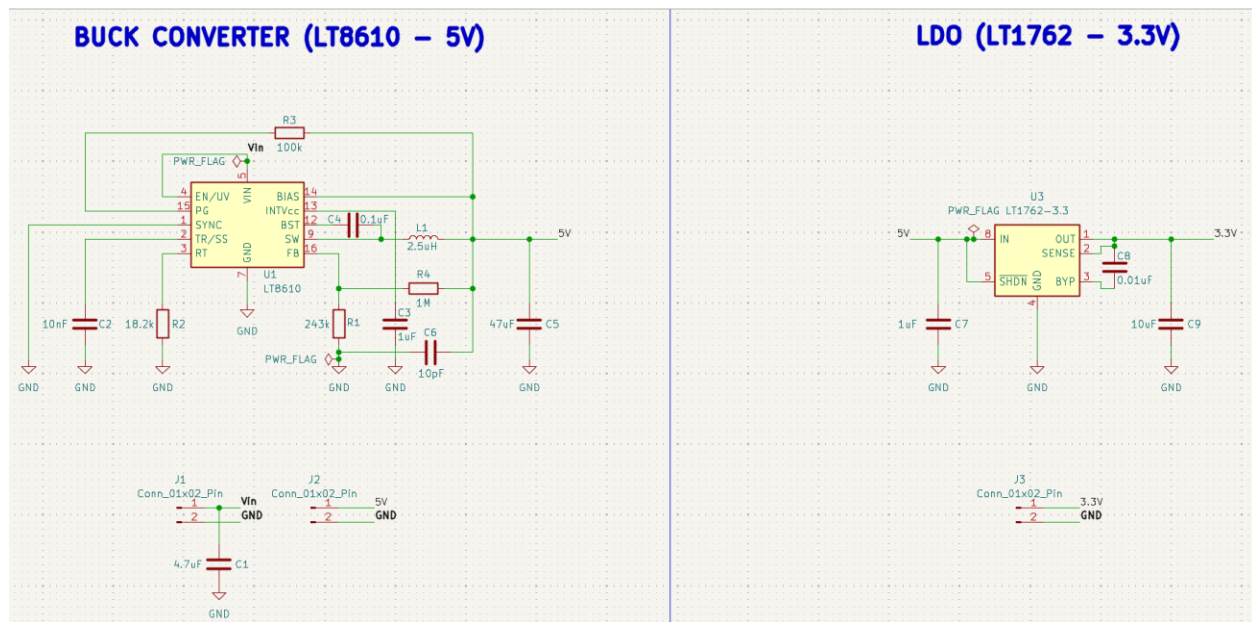


Figure 1.6: Schematic Editor View

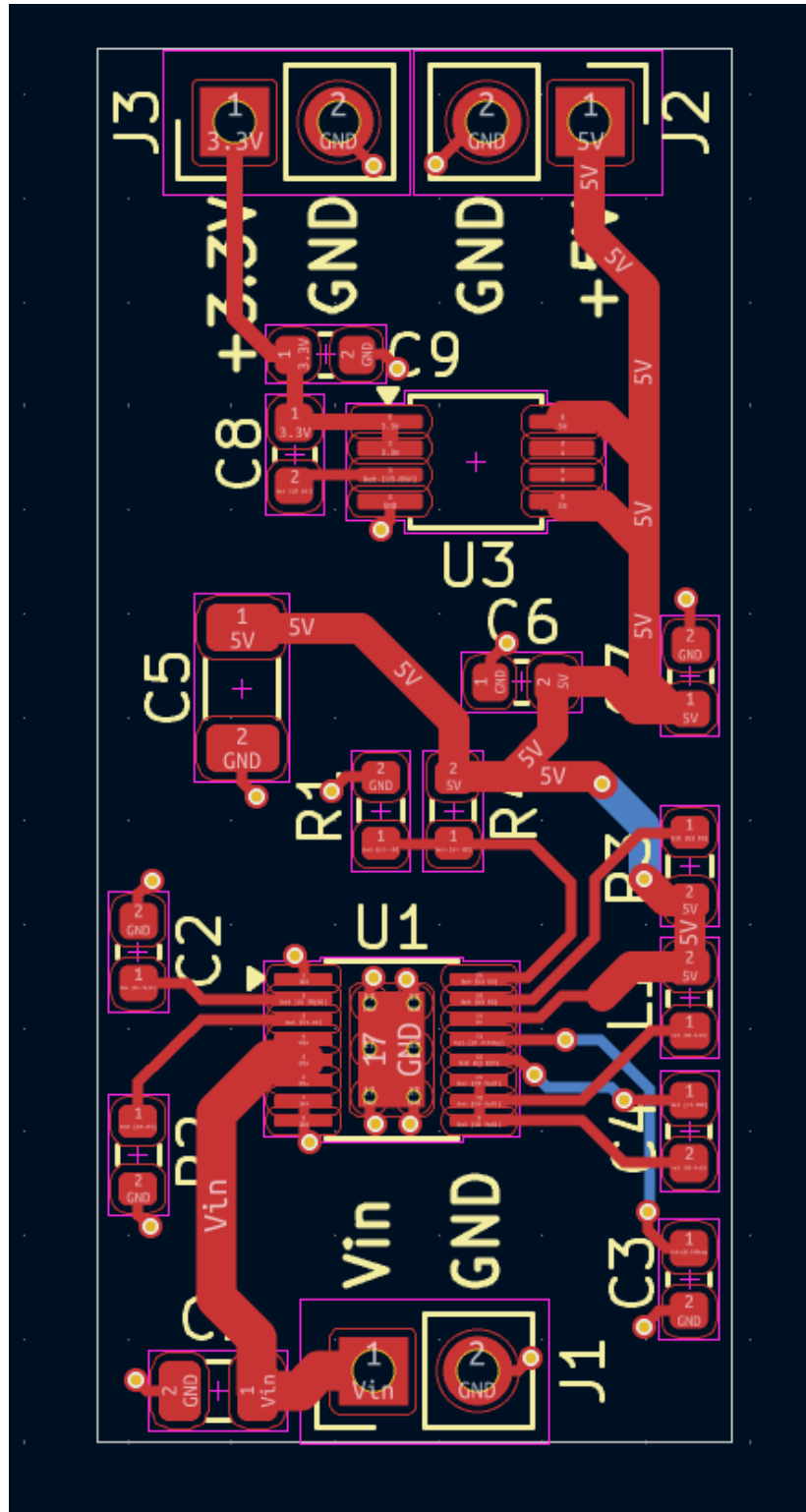


Figure 1.7: PCB Layout Design

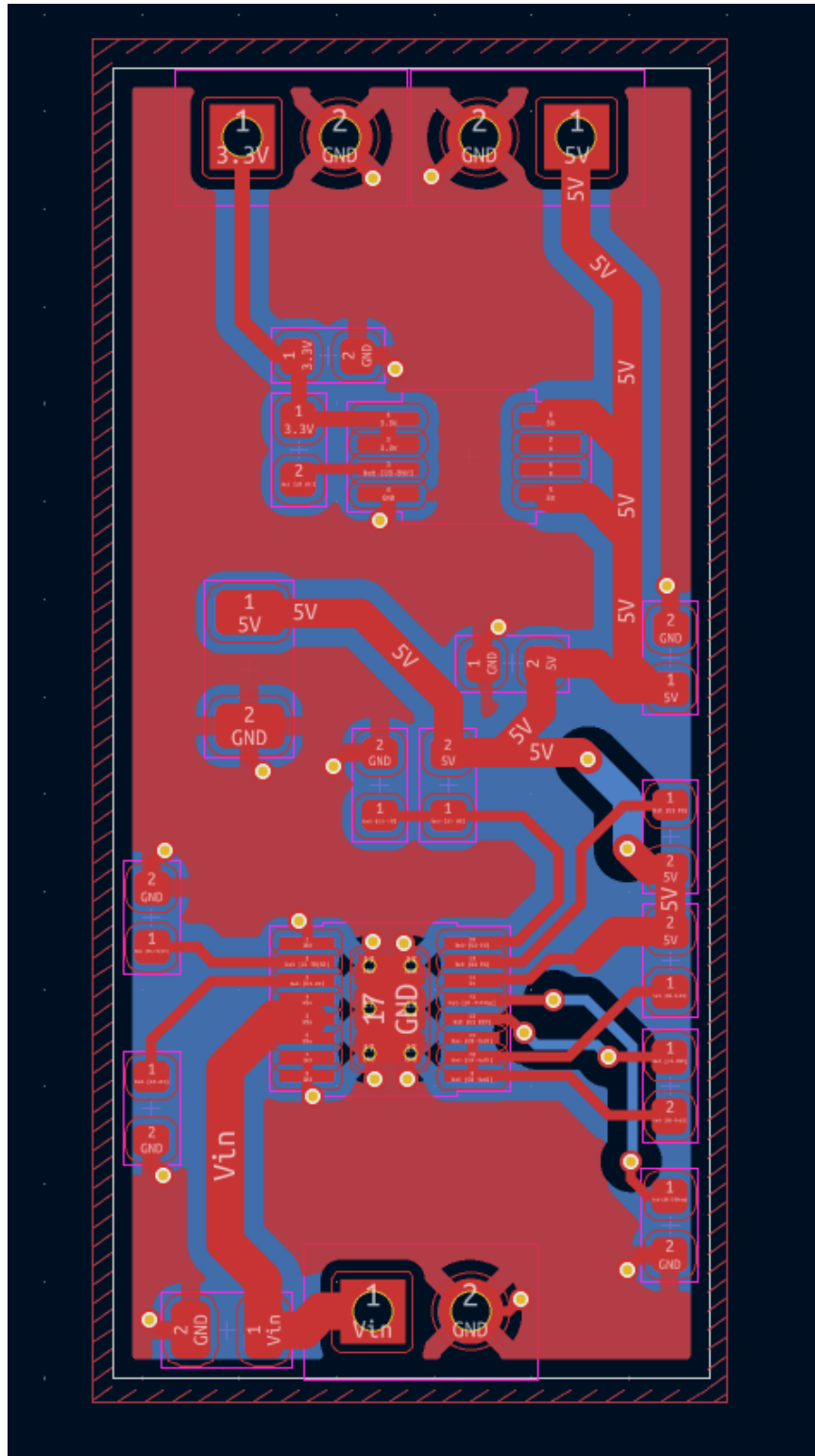


Figure 1.8: PCB Layout Design after copper filling

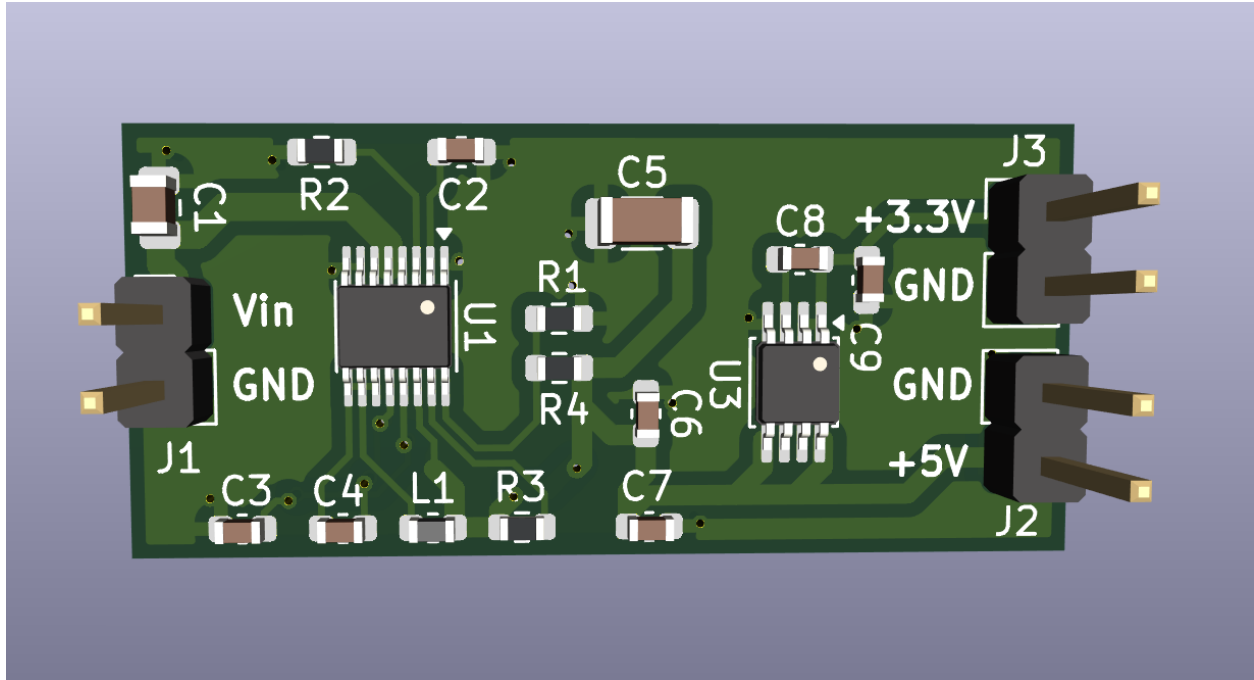


Figure 1.9: Front Layer View

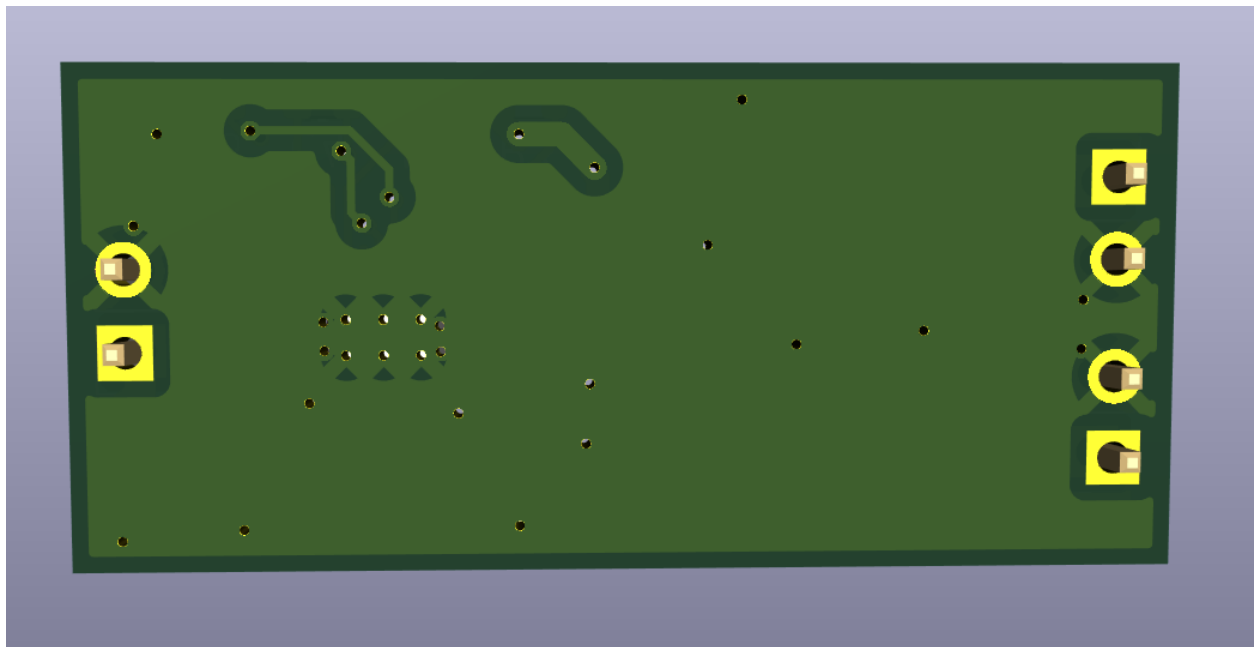


Figure 2: Back Layer View

1.8 Conclusion and Next Steps

Step 2 is complete: both regulation stages have been simulated, debugged, and validated in LTspice, including startup behavior, full battery discharge range, and a realistic sensor load step. The final values in Section 2.7 are ready to carry forward into the PCB schematic in KiCad [done]. Remaining work for this subsystem is limited to real-hardware bench verification once components are assembled, specifically confirming sensor rail ripple remains unaffected while the motor is under load, which cannot be fully modeled in simulation since the motor/ESC sit on a separate, unmodeled high-current rail.